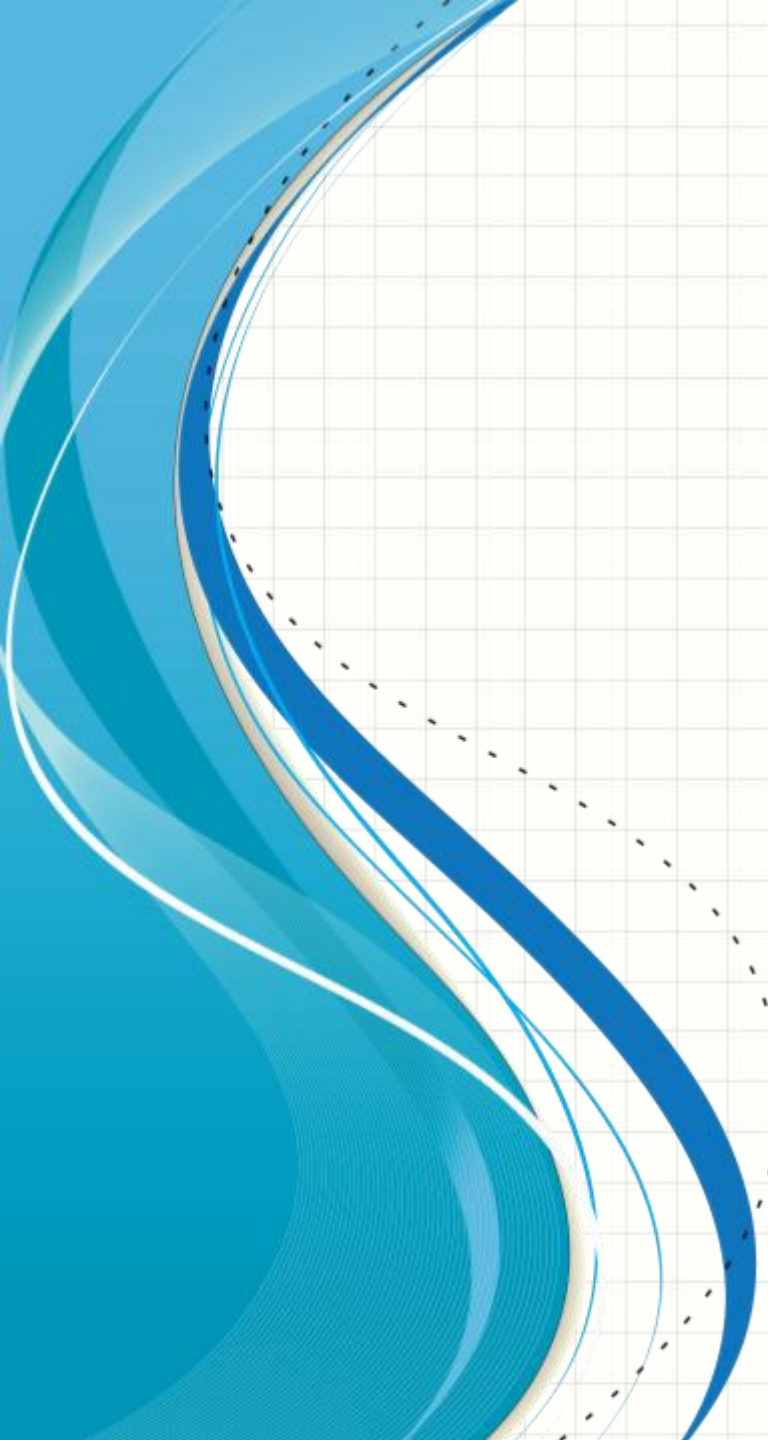




CSE-251

ELECTRONIC CIRCUITS

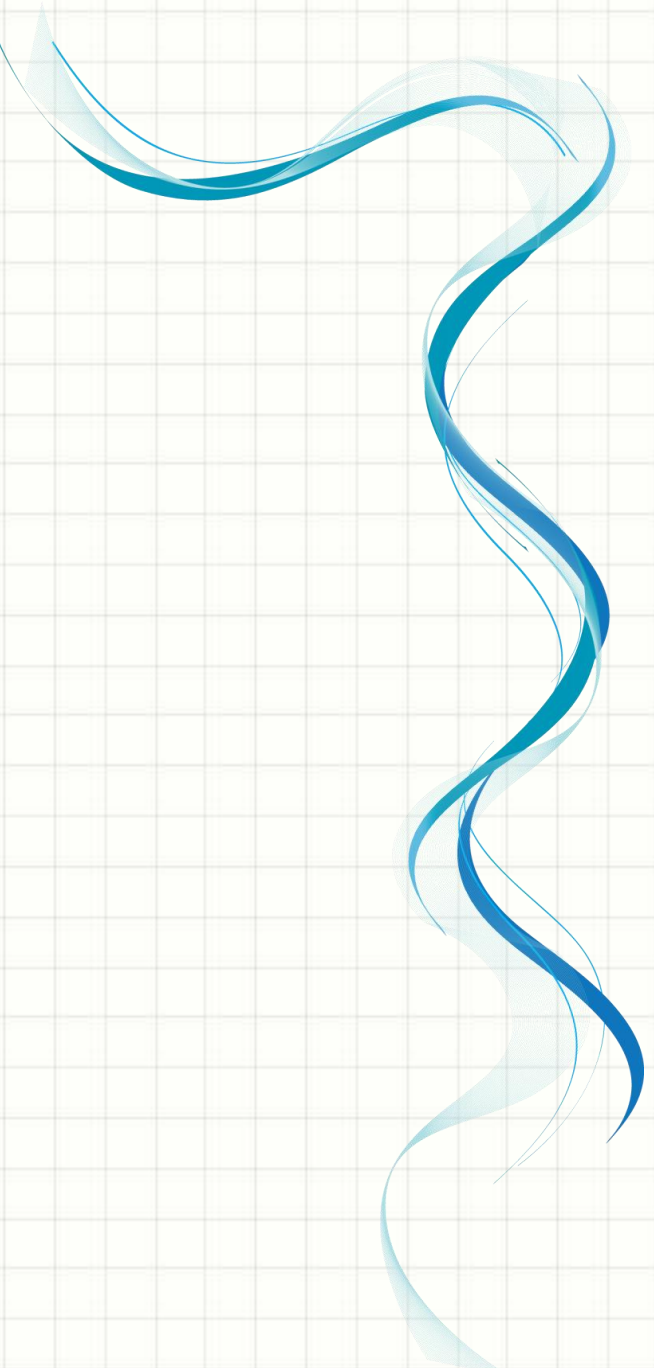
LECTURE-13

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Today's Overview:

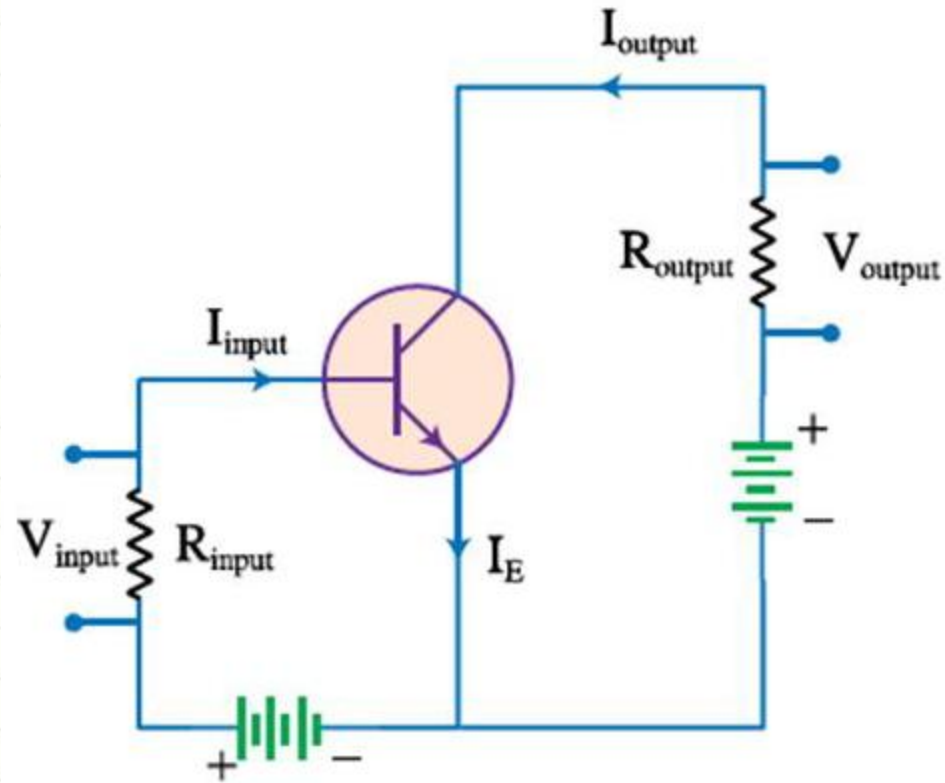
- ❑ *DC Biasing of BJT*
- ❑ *Load Line Analysis*



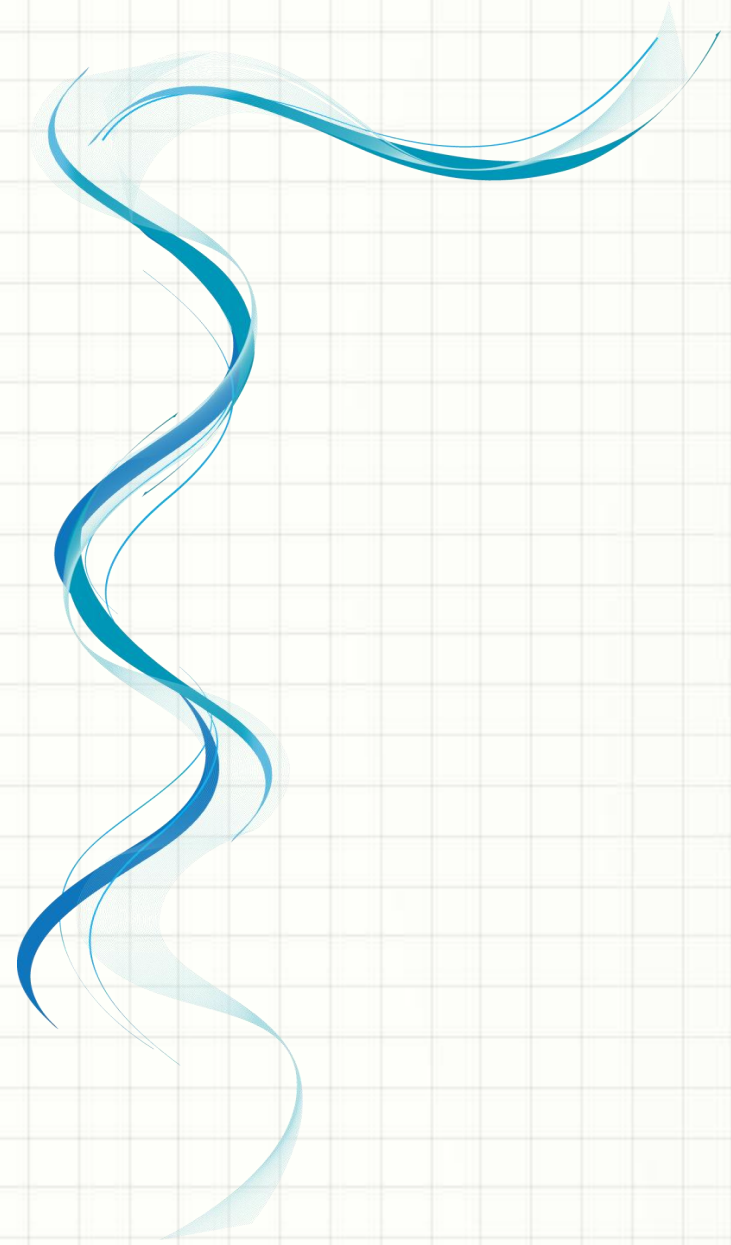
A decorative blue wavy line with a light blue shadow, flowing vertically on the left side of the slide.

DC Biasing of BJT

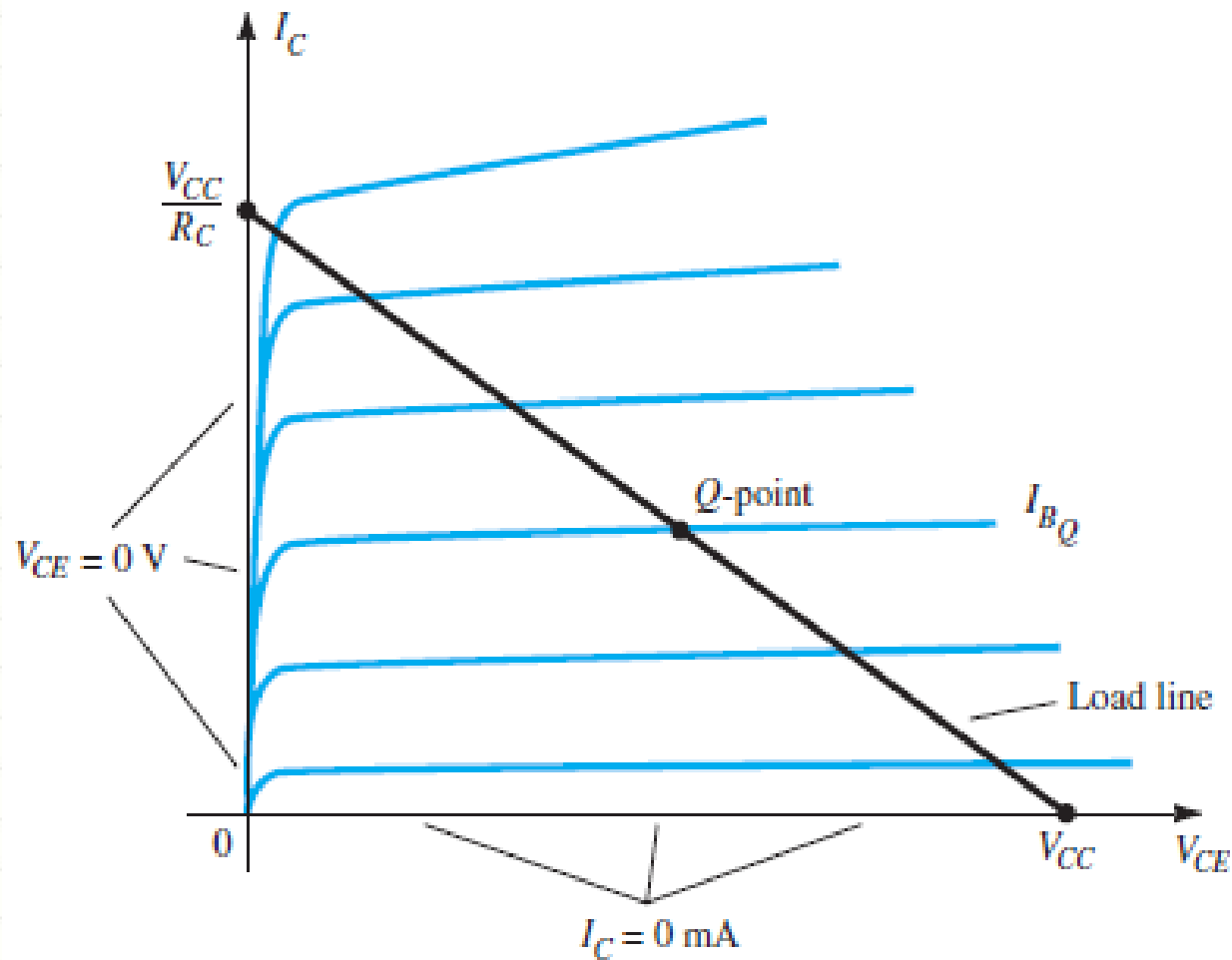
DC Biasing of BJT:

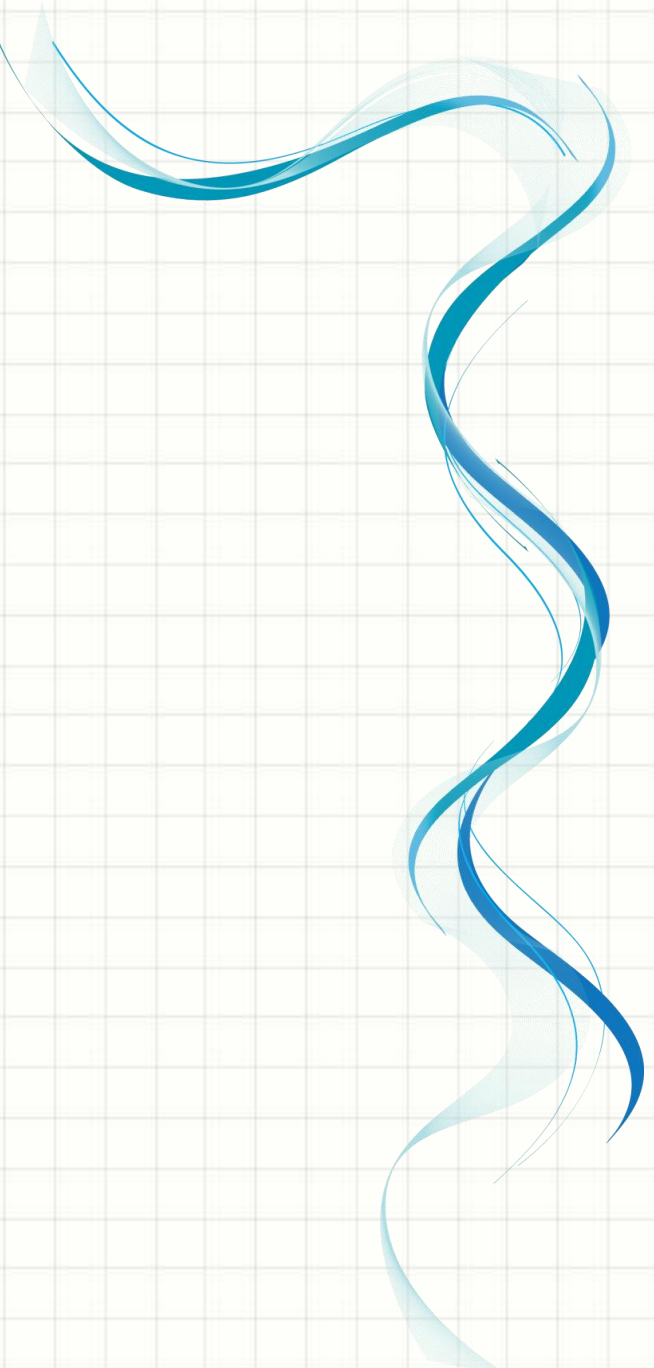


Load Line Analysis



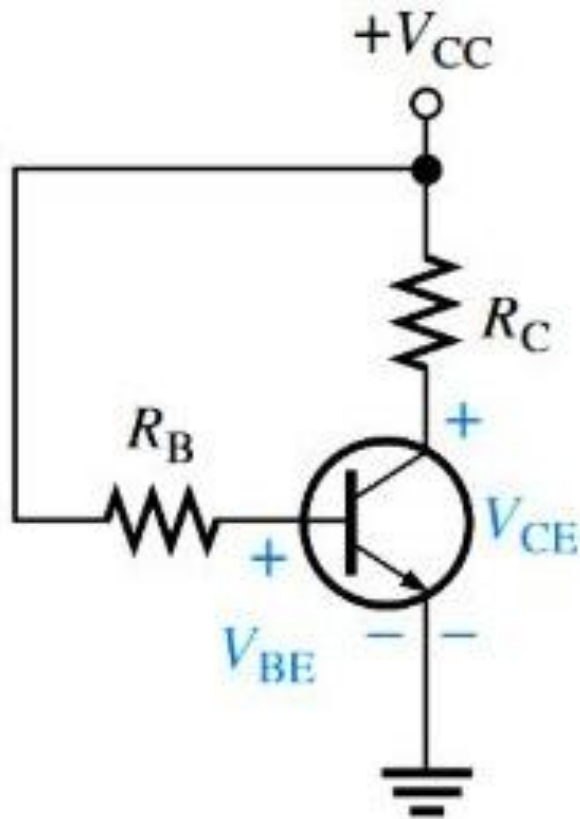
Load Line Analysis:



A decorative blue wavy line with a light blue shadow, flowing vertically on the left side of the slide.

Fixed Bias Configuration

Base Bias:



$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$V_{CE} = V_{CC} - I_C R_C$$

Problem:

EXAMPLE 4.3 Given the load line of Fig. 4.16 and the defined Q -point, determine the required values of V_{CC} , R_C , and R_B for a fixed-bias configuration.

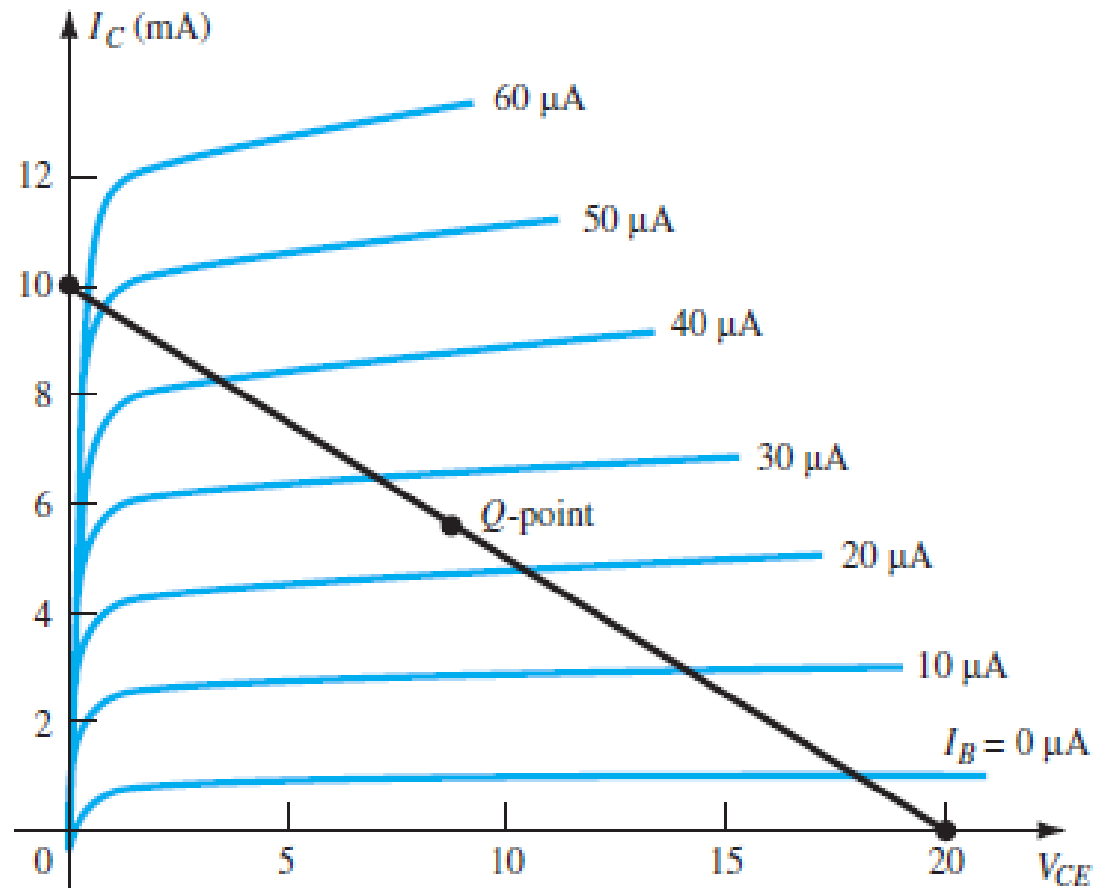


FIG. 4.16
Example 4.3.

Problem:

Solution: From Fig. 4.16,

$$V_{CE} = V_{CC} = 20 \text{ V at } I_C = 0 \text{ mA}$$

$$I_C = \frac{V_{CC}}{R_C} \text{ at } V_{CE} = 0 \text{ V}$$

and

$$R_C = \frac{V_{CC}}{I_C} = \frac{20 \text{ V}}{10 \text{ mA}} = 2 \text{ k}\Omega$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

and

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{20 \text{ V} - 0.7 \text{ V}}{25 \mu\text{A}} = 772 \text{ k}\Omega$$

Problem:

EXAMPLE 4.1 Determine the following for the fixed-bias configuration of Fig. 4.7.

- a. I_{BQ} and I_{CQ} .
- b. V_{CEQ} .
- c. V_B and V_C .
- d. V_{BC} .

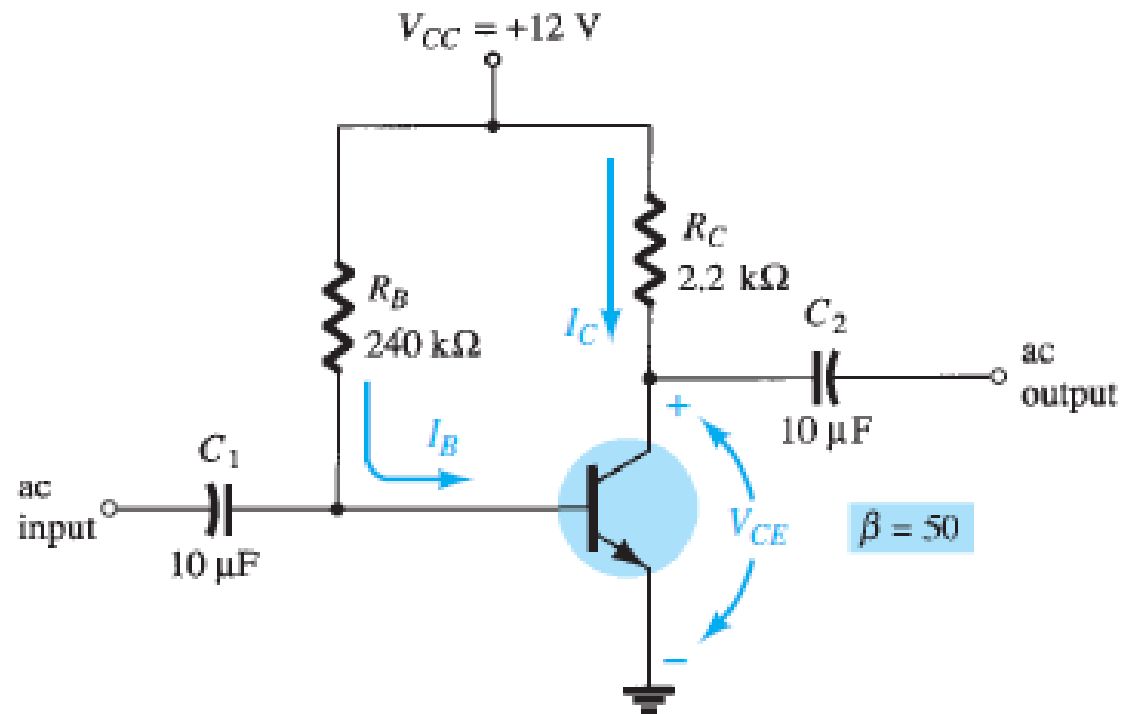


FIG. 4.7

DC fixed-bias circuit for Example 4.1.

Problem:

Solution:

a. Eq. (4.4):
$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\text{ V} - 0.7\text{ V}}{240\text{ k}\Omega} = 47.08\text{ }\mu\text{A}$$

Eq. (4.5):
$$I_{CQ} = \beta I_{BQ} = (50)(47.08\text{ }\mu\text{A}) = 2.35\text{ mA}$$

b. Eq. (4.6):
$$\begin{aligned} V_{CEQ} &= V_{CC} - I_C R_C \\ &= 12\text{ V} - (2.35\text{ mA})(2.2\text{ k}\Omega) \\ &= 6.83\text{ V} \end{aligned}$$

c. $V_B = V_{BE} = 0.7\text{ V}$
 $V_C = V_{CE} = 6.83\text{ V}$

d. Using double-subscript notation yields

$$\begin{aligned} V_{BC} &= V_B - V_C = 0.7\text{ V} - 6.83\text{ V} \\ &= -6.13\text{ V} \end{aligned}$$

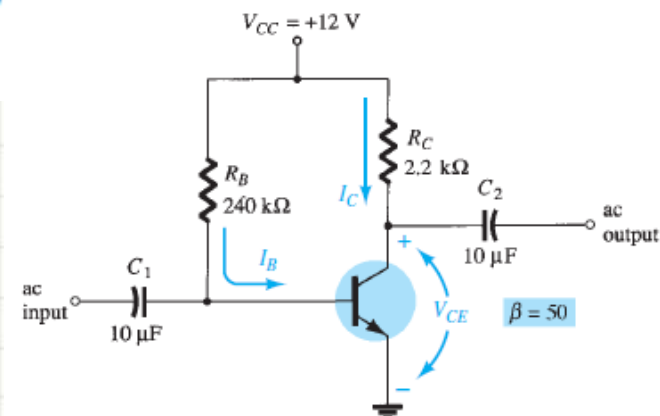


FIG. 4.7

DC fixed-bias circuit for Example 4.1.



QUESTIONS?

SEE YOU IN THE NEXT CLASS

